

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY

Faculty of Engineering

Spatial Heat Mapping of an Electrical Laboratory Tray Dryer

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Criteria	1: Poor (0-20)	2: Fair (30-40)	3: Satisfactory (50-70)	4: Excellent (80-100)	Weight
Problem Definition & Planning	Poorly defined objectives; no clear project plan.	Basic understanding but lacks specific measurable goals.	Clear objectives and a logical sequence of experimental steps.	Highly detailed plan with risk assessment .	15%
Literature Review & Theory	No mention of chemical engineering principles.	Weak link between theory and the specific project.	Good application of theory	Comprehensive review; links project to industrial standards.	15%
Experimental Design & Execution	Major errors in setup; data is unreliable.	Setup is functional but variables are not well controlled.	Systematic approach to testing; data collected is accurate.	Innovative experimental design; excellent troubleshooting.	25%
Data Analysis & Calculations	Significant errors in calculations or data interpretation.	Calculations performed but lack depth (e.g., no heat balance).	Correct use of formulas	Expert analysis; identifies trends and sources of error.	25%
Conclusion & Recommendations	Conclusions do not relate to the original objectives.	Basic conclusions with generic recommendations.	Sound conclusions supported by data; logical next steps.	Strategic recommendations that could be implemented in industry.	10%
Technical Writing & Presentation	Poorly structured; many grammatical/formatting errors.	Readable but lacks professional formatting.	Clear structure; correct use of tables, diagrams, and citations.	Professional engineering report standard; concise and logical.	10%

Executive Summary

This report presents the findings of a spatial heat mapping investigation conducted on an electric laboratory tray dryer. The primary objective was to characterise the internal temperature distribution across a single tray positioned in the upper level of the dryer, operating at a set-point temperature of 60°C.

Temperature measurements were recorded at nine positions arranged in a 3×3 grid on the tray under two conditions: with the internal fan switched off (natural convection) and with the fan switched on (forced convection), the latter data set provided by a concurrent group. The results reveal that both operating modes produce non-uniform temperature profiles across the tray, with cold spots identifiable in all cases.

Under natural convection, temperatures ranged from 45.9°C to 48.9°C, with higher temperatures concentrated in the central positions of the tray. Under forced convection, temperatures ranged from 41.3°C to 50.6°C, indicating a wider spatial variation and a dominant heat gradient from positions near the fan and heating element towards the far end of the tray.

The findings confirm that fan-assisted airflow significantly alters the spatial temperature distribution within the dryer and increases the magnitude of temperature non-uniformity. These results have practical implications for drying operations, where uneven heat distribution leads to inconsistent drying rates and product quality. Recommendations for improved temperature uniformity are discussed.

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Abbreviations and Glossary

°C	Degrees Celsius
PPE	Personal Protective Equipment
3×3	Three-by-three spatial measurement grid
T_{set}	Set-point temperature of the dryer (60°C)
h	Convective heat transfer coefficient (W/m ² ·K)
Q	Rate of heat transfer (W)
Nu	Nusselt number (dimensionless)
Gr	Grashof number (dimensionless)
Pr	Prandtl number (dimensionless)

1. Introduction

1.1 Background and Motivation

Drying is a unit operation of fundamental importance across numerous manufacturing and processing industries. It is commonly the final step in a production process prior to packaging, serving multiple practical purposes: reducing the mass and cost of material transport, improving the handling characteristics of products, providing defined material properties, and preventing moisture-driven corrosion or microbial activity during storage (Coulson J. a., 2002).

The drying process involves the removal of moisture from a solid or semi-solid material, typically through the application of thermal energy. In most industrial drying processes, including tray drying, this is achieved through the vaporisation of liquid water from the surface and interior of the material. The driving force for moisture removal is the difference in vapour pressure between the wet material surface and the surrounding air. Equilibrium moisture content — the point at which a material neither gains nor loses moisture — is governed by the temperature and relative humidity of the surrounding air. (Coulson J. a., 2002)

The quality and efficiency of a drying operation depend strongly on the uniformity of temperature within the dryer. Non-uniform temperature distributions lead to inconsistent drying rates across a tray or batch: regions exposed to lower temperatures dry more slowly, while regions exposed to higher temperatures may suffer product damage such as crystalline structural degradation, shrinkage, or loss of volatile components including flavour compounds (Coulson J. a., 2002). Understanding and mapping the internal thermal environment of a dryer is therefore critical to ensuring product quality and process efficiency. (Coulson J. a., 2002)

1.2 Project Aims and Objectives

This investigation aims to characterise the spatial temperature distribution inside an electric laboratory tray dryer operating at a set-point temperature of 60°C. The specific objectives are:

- To measure and map the 3D temperature profile across the tray surface using a 3×3 measurement grid.
- To investigate the effect of fan operation (forced versus natural convection) on the size and spatial location of cold spots within the dryer.
- To interpret the observed temperature gradients in terms of convective heat transfer mechanisms.

2. Literature Review

2.1 Tray Dryers: Operating Principle

Tray or shelf dryers are widely used for batch drying of granular materials, food products, and individual articles that require controlled drying conditions. In a typical configuration, the material to be dried is spread across a series of trays positioned on racks within an insulated cabinet. Heating is achieved by the circulation of hot air over the material surface. In some designs, the trays may also be heated from below via steam coils, though in the majority of dryers, convective air heating is the primary mechanism (Coulson and Richardson, 2002).

As hot air passes over the wet material surface, it simultaneously increases in temperature and picks up moisture. To maintain drying efficiency, the air may be recirculated and reheated before passing over each successive tray. The relationship between air temperature, humidity, and moisture uptake capacity is described on a psychrometric chart, where the condition of the air moves along an adiabatic cooling line as it absorbs moisture from the material surface.

Modern tray dryers typically consist of a well-insulated cabinet with an integrated fan and heating element. Air velocities of 1 to 10 m/s are common within the drying chamber. Tray areas generally range from 0.3 to 1 m², with material bed depths of 10 to 100 mm depending on the particle size and nature of the product being dried (Coulson J. a., 2002).

2.2 Heat Transfer Mechanisms in Tray Dryers

Heat transfer within a tray dryer occurs primarily through convection, with radiation from the heated walls and element playing a secondary role that is often negligible in low-temperature laboratory-scale dryers. Two distinct convection regimes are relevant to this investigation:

2.2.1 Natural Convection

When the dryer fan is switched off, heat transfer from the heating element to the air and subsequently to the tray surface occurs by natural (free) convection. Natural convection is driven by buoyancy forces arising from density differences in the air due to temperature gradients. Warmer, less dense air rises while cooler, denser air descends, establishing a slow circulatory flow pattern within the enclosed chamber. (Y.A and Ghajar, 2015)

The rate of natural convective heat transfer is characterised by the Grashof number (Gr), which represents the ratio of buoyancy forces to viscous forces, and the Prandtl number (Pr), which characterises the ratio of momentum to thermal diffusivity of the fluid. The product of these two dimensionless numbers — the Rayleigh number — determines the nature and intensity of the natural convection flow. For horizontal heated surfaces facing upward, the Nusselt number (Nu), which relates the convective to conductive heat transfer, can be correlated with the Rayleigh number through established correlations (Y.A and Ghajar, 2015)

In an enclosed dryer with the fan off, natural convection typically produces a temperature distribution characterised by stratification: hotter air accumulates near the top of the chamber while cooler air settles towards the lower regions. Lateral temperature gradients also develop depending on the proximity to the heating element and the geometry of the enclosure. (Y.A and Ghajar, 2015)

2.2.2 Forced Convection

When the fan is operational, forced convection dominates the heat transfer process. A fan drives airflow through the chamber in a defined direction, significantly increasing the convective heat transfer coefficient compared to natural convection. The enhanced air movement promotes more uniform mixing within the chamber, but the flow pattern introduced by the fan also creates localised regions of high and low heat flux depending on the direction and speed of the airstream. (Y.A and Ghajar, 2015) chapter 8

In forced convection, the relevant dimensionless parameter is the Reynolds number (Re), which characterises the nature of the flow (laminar or turbulent). Turbulent flow, which occurs at higher Reynolds numbers, promotes greater mixing and more uniform temperature distribution. However, the spatial arrangement of the fan, heating element, and tray within the dryer cabinet determines whether the forced airflow reaches all regions of the tray uniformly or creates preferential flow paths that result in cold spots (Y.A and Ghajar, 2015).

2.3 Temperature Uniformity and Cold Spots

The concept of a cold spot within a dryer refers to a localised region of the tray or drying chamber that experiences a consistently lower temperature than the set-point or the bulk average. Cold spots arise from any of the following: inadequate airflow circulation to certain regions, heat losses through the dryer walls or door seals, or the shadow effect of trays or structural elements blocking airflow.

From a product quality perspective, cold spots are problematic because they correspond to regions of reduced drying rate. Material in cold spot regions retains more moisture relative to other areas of the tray, leading to non-uniform final moisture content across a batch. In food drying, this can result in spoilage of the wetter regions; in pharmaceutical drying, it can compromise product stability and shelf life.

Spatial temperature mapping — the systematic measurement of temperature at multiple defined points across the tray — is an established technique for identifying and characterising cold spots in drying equipment. It provides a direct, empirical basis for evaluating dryer performance and informing operational decisions such as tray rotation, airflow adjustment, or set-point compensation (Coulson J. a., 2002).

3. Research Methodology

3.1 Equipment

- Electric laboratory tray dryer with internal heating element and fan
- Digital thermocouple (Type K or equivalent) with probe
- Tray positioned at the uppermost level within the dryer
- Ruler and marker for tray position demarcation

3.2 Variables

Independent variable: Fan operating condition (fan off vs. fan on).

Dependent variable: Air temperature at each of the nine defined tray positions.

Control variables: Dryer set-point temperature (60°C), tray level (uppermost), measurement grid configuration (3×3), and thermocouple type.

3.3 Measurement Grid Configuration

A single tray was positioned at the uppermost level of the dryer. The tray surface was divided into a 3×3 grid, defining nine discrete measurement positions. The positions were labelled 1 through 9, with positions 1 to 3 located closest to the heating element and fan, positions 4 to 6 in the middle zone, and positions 7 to 9 at the far end of the tray, furthest from the heating element. This configuration allowed systematic spatial characterisation of temperature across the full tray surface.

3.4 Experimental Procedure

1. The tray dryer was set to a controlled temperature of 60°C and switched on.
2. The dryer was allowed to reach thermal steady state, confirmed when the controller indicated 60°C.
3. The thermocouple probe was positioned at measurement point 1 on the tray grid.
4. The dryer door was briefly opened, the temperature at that position recorded, and the door closed immediately.
5. The dryer was then switched off and the door opened to allow the internal temperature to return to ambient before the next measurement.
6. Steps 1 to 5 were repeated for each of the nine grid positions in sequence.
7. The full procedure was completed with the fan in the off position (natural convection condition).
8. Comparative data with the fan switched on (forced convection condition) was obtained from a concurrent group under equivalent conditions.

3.5 Safety

Throughout the experiment, appropriate personal protective equipment was worn at all times. Laboratory coats, closed shoes, and long trousers were worn to minimise skin exposure to the heated surfaces and the warm air released when the dryer door was opened. Heat-resistant gloves were used when handling the tray and probe near the dryer interior. All equipment was operated in accordance with laboratory safety protocols, and the dryer was not left unattended while at operating temperature.

4. Results and Discussion

4.1 Measured Temperature Data

The temperatures recorded at each of the nine tray positions under both operating conditions are presented in Tables 4.1 and 4.2 below. The dryer set-point temperature was 60°C in both cases.

Table 4.1 — Measured tray temperatures under natural convection (fan off)

Position	Zone	Temperature (°C)	Element Set-Point (°C)	Fan
1	Near element	46.3	60	Off
2	Near element	47.8	60	Off
3	Near element	48.6	60	Off
4	Middle	48.7	60	Off
5	Middle	48.9	60	Off
6	Middle	48.2	60	Off
7	Far end	46.6	60	Off
8	Far end	45.9	60	Off
9	Far end	46.4	60	Off

Table 4.2 — Measured tray temperatures under forced convection (fan on)

Position	Zone	Temperature (°C)	Element Set-Point (°C)	Fan
1	Near element/fan	50.6	60	On
2	Near element/fan	48.5	60	On
3	Near element/fan	45.1	60	On
4	Middle	44.3	60	On
5	Middle	41.3	60	On

6	Middle	41.3	60	On
7	Far end	43.3	60	On
8	Far end	45.0	60	On
9	Far end	44.5	60	On

[Figure 4.1 and Figure 4.2 — Heatmap visualisations to be inserted here from Python/matplotlib output]

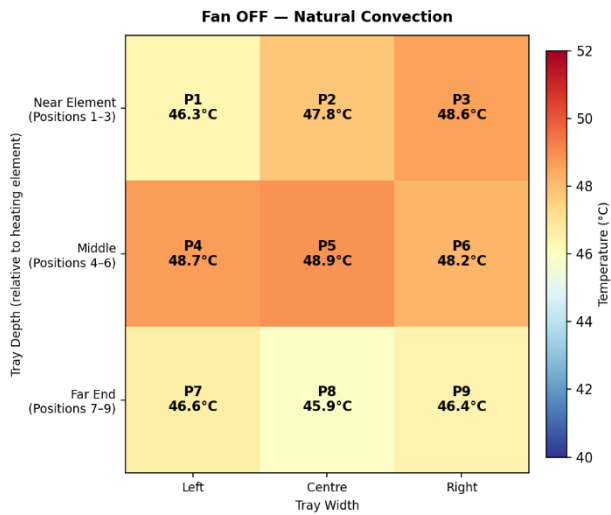


Figure 1 spatial heat map of tray fan off

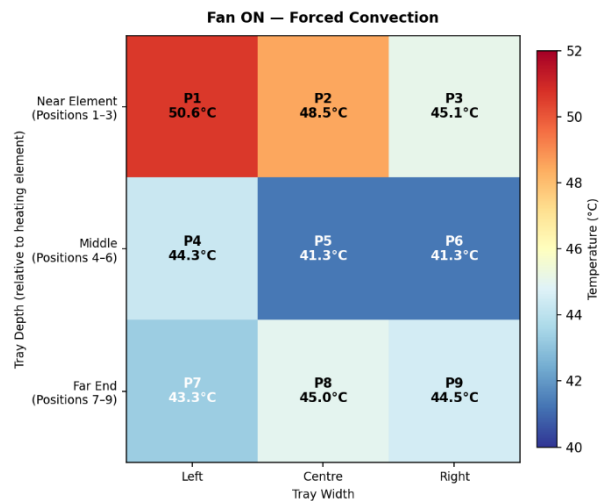


Figure 3 spatial heat map of tray fan on

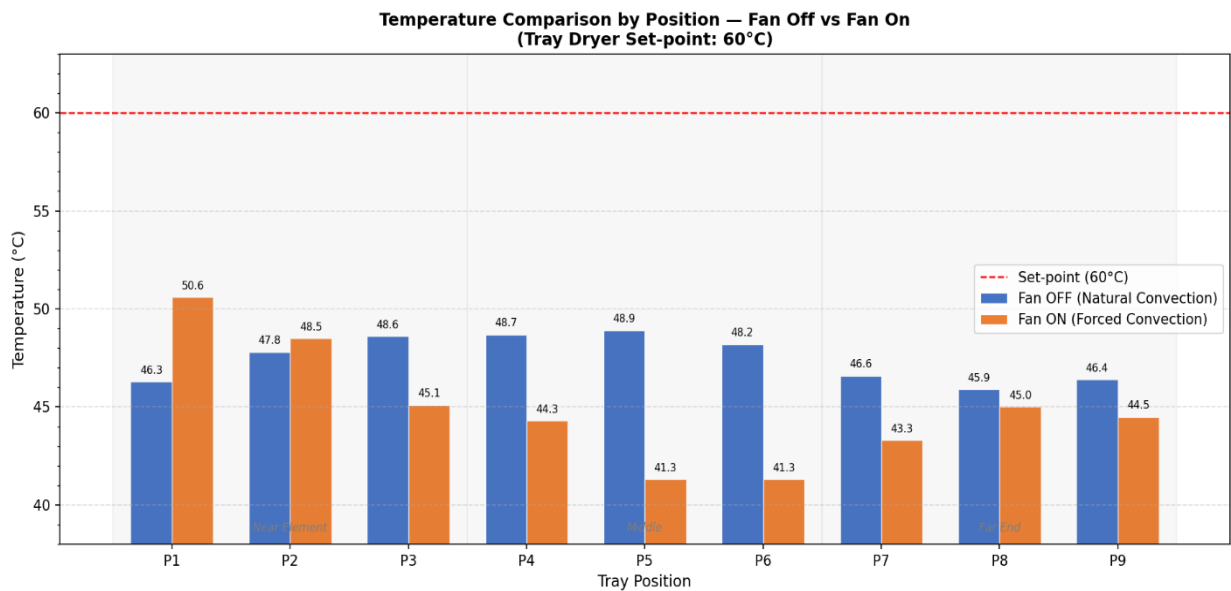


Figure 2 bar chart comparison

4.2 Natural Convection Analysis (Fan Off)

Under natural convection conditions, the temperature range across the tray was 45.9°C to 48.9°C, representing a maximum spatial variation of 3.0°C relative to a set-point of 60°C. This variation, while relatively modest in absolute terms, is significant given that all positions register temperatures considerably below the set-point, indicating a substantial temperature deficit between the heating element and the air measured at tray level.

The temperature distribution shows a characteristic pattern: the middle zone positions (4, 5, 6) record the highest temperatures, while the near-element zone (positions 1 to 3) and the far-end zone (positions 7 to 9) register lower temperatures. This is consistent with natural convection behaviour in an enclosed heated space. Hot air rising from the heating element migrates towards the centre of the tray as buoyancy-driven circulation establishes itself within the chamber. Positions closest to the element may paradoxically register slightly lower temperatures than the central zone if the rising plume of heated air detaches from the element surface and convects preferentially to the central region before spreading.

The coldest position was position 8 (45.9°C), located at the far end of the tray. This is consistent with heat loss to the dryer walls and the reduced intensity of natural convection flow at the periphery of the enclosure. Under purely diffusive and weakly convective conditions, the far end of the tray is least exposed to the primary heat source, making it the most likely location for cold spots.

4.3 Forced Convection Analysis (Fan On)

With the fan operating, the temperature profile changes substantially. The overall temperature range increases from 3.0°C (fan off) to 9.3°C (fan on), spanning from a minimum of 41.3°C at positions 5 and 6 to a maximum of 50.6°C at position 1. This wider spread indicates that forced convection, while enhancing heat transfer at positions directly exposed to the airstream, simultaneously creates more pronounced cold spots in regions of poor airflow.

The highest temperature is recorded at position 1, the point closest to both the fan and the heating element. This is expected: the fan directs heated air along a defined flow path, and the first positions encountered by this airstream receive the greatest heat flux. As the air travels across the tray towards the far end, it progressively loses thermal energy to the material and tray surface, resulting in a declining temperature gradient in the primary flow direction.

The coldest positions (41.3°C) are positions 5 and 6, situated in the middle zone of the tray. This is unexpected if a simple linear gradient from positions 1 to 9 is assumed. The pattern suggests that the fan-driven airflow does not traverse the tray in a simple linear path. Instead, the jet of air from the fan may attach to the near-element side of the tray, then partially separate

or recirculate, creating a low-velocity wake region in the middle zone. Positions 7 to 9 at the far end show a slight temperature recovery (43.3°C to 45.0°C), possibly due to recirculation of air within the chamber that brings relatively warmer air back towards the far end.

4.4 Comparison of Convection Modes

Comparing the two conditions, natural convection produces a more uniform temperature distribution with smaller spatial variation (3.0°C range) but a consistently larger deficit from the set-point. Forced convection raises the peak temperature closer to the set-point at the most favourable position but introduces a more complex and uneven temperature field with larger cold spots.

For drying applications, neither condition is ideal. Natural convection leads to slow, relatively uniform but energy-inefficient drying. Forced convection increases the average heat flux and drying rate but risks over-drying material in high-temperature zones while under-drying material in cold spots. These findings underscore the importance of understanding the spatial temperature environment of the dryer when designing or evaluating drying protocols.

4.5 Sources of Error and Limitations

Several sources of error are acknowledged in this investigation. Opening the dryer door to position the thermocouple at each measurement point inevitably disturbs the internal thermal environment, causing a transient temperature drop at the measurement location. Although the dryer was allowed to return to steady state between measurements, small residual thermal disturbances cannot be excluded.

The thermocouple probe was positioned manually at each grid point, introducing potential positional variability between measurements. The tray was also at the uppermost level, meaning results are specific to that tray position and may not be representative of lower tray levels where airflow patterns and proximity to the heating element differ.

Finally, the fan-on data was sourced from a separate experimental group rather than repeated under identical conditions by the same group. Minor differences in experimental procedure, thermocouple calibration, or dryer warm-up protocol between groups may introduce systematic differences in the comparative data set.

A heat balance to determine the thermal efficiency of the dryer could not be completed in this investigation. The electrical power rating of the dryer unit was not available, and ambient room temperature was not recorded during the experimental session. Without these values, the heat

input and heat losses cannot be quantified. This is acknowledged as a limitation of the study, and it is recommended that future investigations record ambient conditions and obtain the equipment power rating prior to experimentation.

5. Conclusion and Recommendations

5.1 Conclusions

This investigation successfully characterised the spatial temperature distribution within an electric laboratory tray dryer operating at a set-point of 60°C, under both natural and forced convection conditions. The following conclusions are drawn:

- Non-uniform temperature distributions exist across the tray surface under both operating modes, confirming the presence of cold spots in all conditions tested.
- Under natural convection (fan off), the temperature variation across the 3×3 measurement grid was 3.0°C, with the highest temperatures in the central zone and coldest temperatures at the far end of the tray.
- Under forced convection (fan on), the temperature variation increased significantly to 9.3°C. The highest temperature was recorded at the position closest to the fan and heating element, while the most pronounced cold spots were located in the middle zone of the tray.
- Fan operation shifts the dominant heat transfer mechanism from buoyancy-driven natural convection to directed forced convection, fundamentally altering the spatial temperature field. This increases peak tray temperatures but also increases the severity of cold spots.
- The consistent temperature deficit between the dryer set-point (60°C) and the measured tray temperatures in both conditions indicates heat losses within the dryer system and suggests that the thermostat set-point alone is not a reliable indicator of the temperature experienced by material on the tray.

5.2 Recommendations

- Tray rotation during drying operations should be implemented to compensate for the non-uniform temperature field. Rotating trays at defined intervals ensures more uniform exposure of all material to the available heat.
- For applications requiring tight moisture uniformity, fan-assisted operation should be combined with careful consideration of tray loading and position. Material should not be concentrated in regions identified as cold spots.

- Future investigations should repeat measurements at multiple tray levels and with the fan at variable speed settings to build a more complete characterisation of the dryer's thermal performance.
- A heat balance calculation incorporating the measured tray temperatures, the element power rating, and ambient conditions would allow quantification of the dryer thermal efficiency and should be included in subsequent work.
- The use of multiple simultaneous thermocouples placed at all nine positions, rather than sequential single-point measurement, would eliminate the thermal disturbance introduced by repeated door opening and provide a more accurate steady-state temperature map.

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